# INTERIOR DOCTRINE — SPACE HANGAR

Everything interiors need: room fit, decoupled MoGe normals, a self-checking semantic mask, and the patch loop.

FILE **atlas_interior_hangar_workflow.json** VERIFIED **2026-07-16**

## THE STORY

Interiors break the outdoor assumptions: there is no sky (so the sky heuristic must prove itself harmless), depth models behave differently in enclosed spaces (MoGe is the interior specialist), and a Manhattan room fit (AtlasDeriveInteriorRoom) can carry flat walls where relief noise would wobble. This workflow runs all of it on an ACEScg hangar plate.

Two details are the teaching moments. First, AtlasMogeNormals DECOUPLES relight normals from the depth source: you keep the depth model you trust for geometry while a MoGe normal pass attaches cleaner per-pixel normals for the viewport's relight — wired between AtlasDepthMap and the relight layer. Second, AtlasSemanticMask prompted 'sky' returns ~0% on an interior — a deliberate self-check proving the mask machinery works before you trust it on scope masks.

The 📐-gated patch loop ships wired but paused: orbit the viewport to a reveal, click ⬛ Render Passes then 📐 Extract Angle, and the re-queue feeds the projected render back as a patch at the EXACT measured pose, painting only where the primary camera can't see. With a Qwen Multiple-Angles LoRA you'd generate the patch image instead — same wiring.

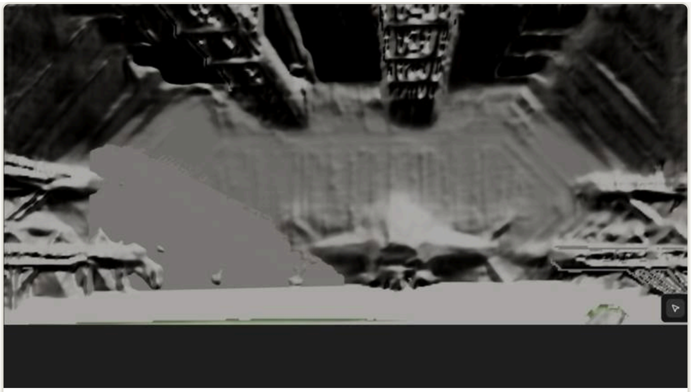
## IN THE VIEWPORT — PROJECTED VS. THE GREY MESH

SOURCE PLATE

spacehangar_32bit_acescg.exr · ACEScg float — the single still photograph everything below is reconstructed from.

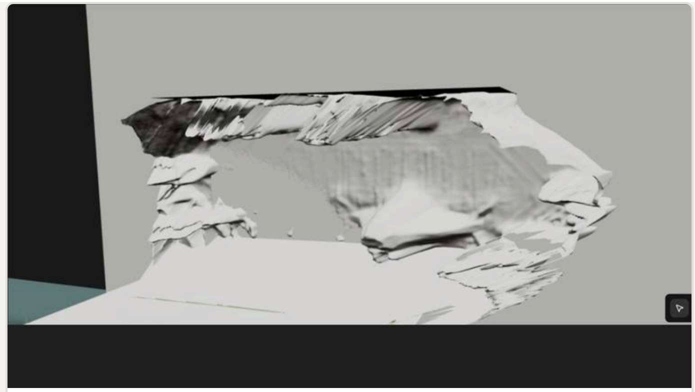
PROJECT ON

The photo reassembled on the recovered camera — texels assigned by ray, perfect from this view.



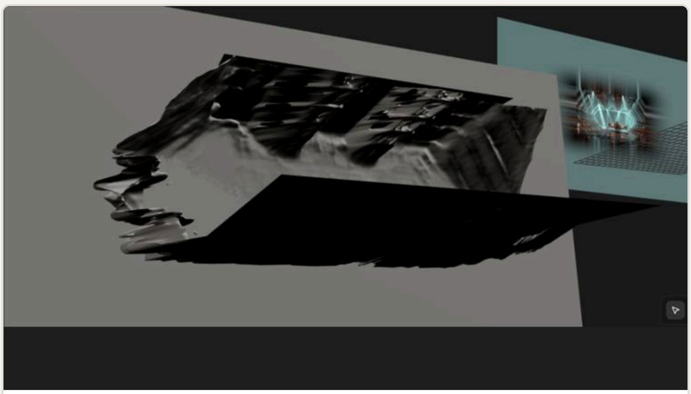
**PROJECT OFF**

The same geometry as a grey mesh, recovered view — what the depth-derived surfaces actually are.



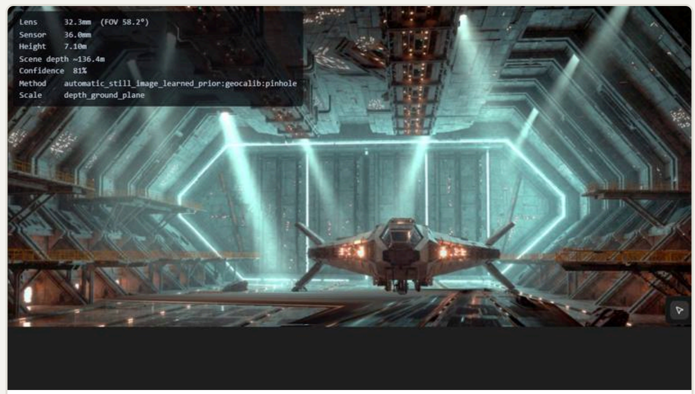
**GREY · ORBIT LEFT**

Orbited off-axis: the reconstructed cone. Coverage ends where the camera never saw.



**GREY · ORBIT RIGHT**

The opposite angle — silhouette tears read as deliberate holes, not errors.




**INFO HUD**

ℹ overlay: solved lens, sensor, camera height, confidence — the solve's numbers in the corner.

## NODES ON STAGE

`OCIORead` `AtlasAssessImage` `AtlasSolveGate` `AtlasDepthMap` `AtlasMogeNormals` `AtlasSemanticMask` `AtlasDeriveInteriorRoom` `AtlasDeriveReliefMesh` `AtlasMergeGeometry` `AtlasCleanPlateLayer` `AtlasOcclusionMask` `AtlasAddPatchView` `AtlasExportMayaLayers`

## RETUNE ON YOUR PLATE

- **Two gates.** ▶ Continue on the VLM (automatic here), then ✅ Approve Solve. A 2-second run means the gate is closed, not broken.
- **Depth model.** V2-Indoor ↔ MoGe to taste — MoGe is often cleaner on enclosed shots; its far-field runaway only bites outdoors.
- **The patch branch pauses BY DESIGN.** It stays silent until 📐 Extract Angle produces a measured pose. The HUD announces it; nothing is hung.

- **normal_model on AtlasMogeNormals.** vitl best · vitb lighter GPU · vits is CPU/MPS-viable.

## RUN-VERIFIED

**RUN ✓** The relight layer carries the MoGe normal map (verified in the viewport payload); semantic 'sky' self-check ≈0% as designed; patch branch correctly paused; .ma + per-layer assets written.

## REQUIREMENTS

`OCIO/EXR env · [moge] · VLM (lmstudio) · ComfyUI-RMBG`